

DREAM MR: Distraction Regulation for Effective Attention Management using Mixed Reality

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ABSTRACT

Public places can be overstimulating, making focusing on a task difficult. This can be particularly problematic to people in the neurodivergent spectrum. This paper presents the initial plan to develop DREAM MR, an assistive framework to regulate the amount of distractions a person perceives from the environment. The objective of the framework is to temporarily and seamlessly attenuate elements in the vicinity of a workspace that have the potential to distract users from their current task. For each object, the amount of attenuation to apply over time will be determined by calculating three criteria: potential for distraction and overstimulation, relevance to the task, and ability to promote connectedness to the environment. DREAM MR will achieve this by leveraging recent advancements in Mixed Reality such as real-time environment tracking and understanding and video pass-through. This framework will promote personal accessibility and functionality in public spaces for all individuals.

Index Terms: Human-centered computing [Human computer interaction (HCI)]; Interaction paradigms—Mixed / augmented reality; Human-centered computing [Accessibility]; Accessibility technologies Computing methodologies [Artificial intelligence]; Computer vision—Computer vision tasks

1 INTRODUCTION

People can increase their productivity and motivation by changing their workspace [12]. Although dedicated co-working spaces have become more prevalent after the COVID-19 pandemic [9], people are often seen working on their laptops in public environments such as cafes and libraries [7]. However, these environments are often filled with elements that can distract or overstimulate individuals, which is ultimately detrimental to their productivity efforts [27, 18]. This overstimulation can be particularly harmful for neurodivergent individuals, who can be more prone to experience sensory overload due to the excessive rate or intensity of the elements in the environment [6]. Creating a balance between positive stimulation and minimal distraction is difficult and often infringes on the original purpose of the space [28].

This research will bridge this gap by introducing DREAM MR, a Mixed Reality framework to regulate the amount of distractions a person perceives from the environment. DREAM MR's objective is to temporarily and seamlessly attenuate elements in the vicinity of a workspace that have the potential to distract users from their current task. These attenuations can be in the form of blurring the elements, changing their opacity, or modifying their colors. For each object, these attenuations can change over time according to three criteria: potential for distraction and overstimulation, relevance to the task, and ability to promote connectedness to the environment. The

main goal of this research is to promote personal accessibility and functionality in public spaces for all individuals.

2 LITERATURE REVIEW

Studies using Augmented Reality (AR) to remove distractions and focus attention have been conducted with both neurotypical [3, 26, 25] and neurodivergent individuals [11, 32, 19, 8, 35]. These studies often involve the use of Diminished Reality, a technique that removes elements from the real environment from sight as opposed to adding virtual elements. Often times, these approaches work by separating objects in the scene into relevant or non-relevant to the task. This works well in controlled environments, but can fail in more "in-the-wild" environments such as walking or people in cafes [10].

Recent improvements in artificial intelligence and mixed reality allow for more complex programs that are able to handle "in-the-wild" environments. Artificial intelligence, particularly environment understanding, has become far more accurate [29] and now has the potential be utilized in effectively making programs adaptive to a variety of spaces. Similarly, mixed reality pass-through has undergone significant improvement [34]. Pass-through allows for more control than augmented reality due to the entire field of view being filtered through cameras. DREAM MR aims to utilize these advancements to create an adaptive framework able to handle every aspect of the environment and create personalized spaces for the user.

3 OVERVIEW OF DREAM MR

The DREAM MR framework will develop a method of removing or modifying elements in a dynamic environment to suit the user. This framework can be broken into seven major modules. To ensure that the framework can be utilized in any public situation, it must first understand both the environment and the purpose of the user.

If the user enters a cafe to do work, as in Figure 1, DREAM MR must utilize environment understanding and task understanding to get the basic information needed to assess which modifications to make. The framework follows with the **Environment and Task Understanding Module**, which will utilize dynamic environment understanding, such as YOLO [30] or the Segment Anything Model [22] to isolate objects in the environment for further analysis. Similarly, task understanding can be intuited by utilizing human action understanding models [1, 21, 33, 37]. These models can be used to establish the task being undertaken (Fig. 1a).

Secondly, the **Sectioning Module** separates the space into the user's foveal view, peripheral view, and the space outside of the user's view. This is done to emulate how the brain processes visual stimuli from each of these areas [25, 36]. The space is also segmented into workspace and non-workspace. This can be done similarly to a current method of workspace determination in a group [17], where the space and workers are analyzed with environment and action understanding to create areas of interest. Instead of analyzing workers, this could be done through grouping the users fixation points in the environment over time. These segmented spaces will be used to classify objects for analysis in the next modules.

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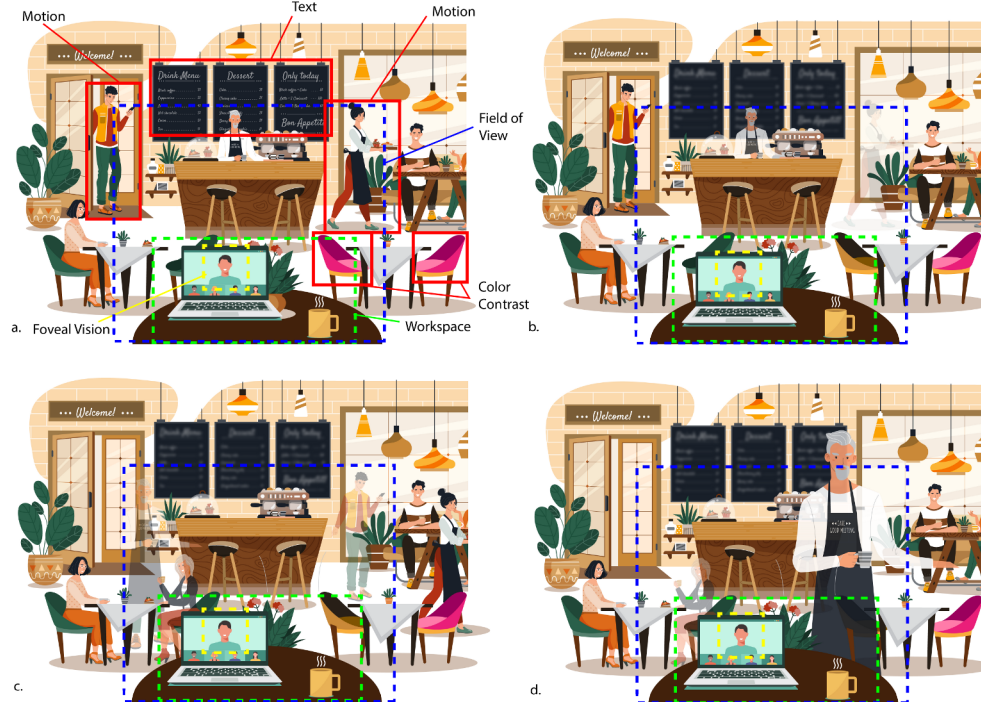


Figure 1: A. This is a simplified identification of potential distractions with only the primary attribute considered. This determines what will be altered and how. B. The initial modifications based on A. The movement in the field of view has been removed, and the high color contrast present in the workspace has been altered. The text on the blackboard has been blurred to be unreadable. C. Nearby people in the field of view are reduced in opacity proportionate to the distance they are from the user. D. As the figure continues to move forward into the workspace, he continues to increase in opacity due to the likelihood of the figure interacting with the user.

The third part of the DREAM MR framework involves calculating two main considerations for each object: *index of distraction* and *index of relevance*. Two independent modules, each working in parallel, are developed to tackle these considerations. The **Distraction and Overstimulation Calculation Module** will analyze the attributes of the object as is shown in Figure 1a. This can be done through common computer vision routines that detect motion, excessive color contrast with the environment, and text; all known sources of user distraction [24] and possible causes of overstimulation in neurodivergent individuals [4]. Similarly, the **Relevance Module**, will analyze the relevance of the object to the task itself. This can be affected by the task understanding, environment understanding, and its relative positioning. For example, if the task is to read and respond to emails, an object that is identified as a laptop in the workspace is likely to be relevant to this task. This can be achieved through the previous use of environment understanding along with a machine learning algorithm trained to understand the use of objects in a variety of tasks with assigned weights for relevance to certain tasks [23].

Afterwards, the **Attenuation Calculation Module** receives the values of the previous three modules to determine which objects need to be attenuated. This balance can be seen in Figures 1b-d, where the location and status of the objects changes how they are handled. This may be done through an algorithm such as Ripple Down Rules [13] to establish flexible rules for decision making based on the indices determined by the previous modules.

Finally, the **Environment Attenuation Module** will perform the attenuations identified previously with a combination of video pass-through routines and diminished reality methods, e.g., blurring aspects of the object, or changing the color of the object [15], as shown in Fig. 1b-d. These modifications will be done with the

express intent of reducing stimulation from the object.

DREAM MR goes through these modules for all objects in the environment periodically to dynamically alter the space to promote focus and comfort for neurodivergent individuals. Distractions and other excessive stimulation will be mitigated by removing or altering the attributes of select objects without compromising the task being undertaken by the user.

4 FUTURE WORK AND CHALLENGES

DREAM MR confronts a number of challenges the frequently occur within dynamic frameworks. Overall, the complexity of each module, particularly environment understanding and modifying the visuals, may affect efficiency due to limited space and power in the currently available HMDs. Environment understanding is limited by the size of the machine learning model. A small model can fit within the limited computational space of a local HMD, but may not be trained for all possible elements in an environment [38]. This may promote the need for a GPU server to handle the video effectively, bringing in external hardware and adding complexity to the system. Similarly, task understanding requires complex implementation and may not be capable of full automation in its current stages. This severely limits the capabilities of the later analysis and generalizability to a wide variety of environments. With this, there are also privacy and security concerns in recording the environment for analysis, such as filming private information and non-consenting individuals, that will need consideration before utilizing the framework in public. Furthermore, the three main considerations on which the framework is based are largely subjective. This could mean that each individual's experience with the framework will vary. As such, effectively adapting DREAM MR over time will be a challenge [5].

5 CONCLUSION

DREAM MR is designed to be a framework that mitigates distractions and overstimulation for an individual in public spaces. This is done through real time environment understanding and task understanding to determine the potential each object in an environment has to cause distraction and overstimulation, the relevance the object has to the task, and how the object influences connectedness with the environment. This information is consolidated to determine visual modifications to each object in the space to promote user focus and comfort without detaching the user from the environment entirely. DREAM MR will reveal important considerations to create more accessible MR interfaces for everyone.

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